The Geometry of Mobile Lethality

A Large-N Theory of
Submarine Attrition, Encounter Processes, and Strategic Survival

Soumadeep Ghosh with ChatGPT (OpenAI)

Kolkata, India

Abstract

This paper develops a mathematical theory of submarine attrition in a mobile battlespace. Consider N submarines moving from origins to destinations along straight-line trajectories. Each submarine possesses a circular destruction radius and destroys any adversary entering that radius. The resulting system is formulated as a dynamic geometric encounter process, a directed kill graph, and a large-N stochastic field model.

We derive exact pairwise interception conditions, construct an event-driven elimination algorithm, establish a Poisson encounter approximation, and obtain a closed-form asymptotic expression for the survivor fraction. Monte Carlo simulations reveal exponential survival decay as a function of destruction radius and force density. Connections are drawn to kinetic theory, warfare economics, deterrence theory, naval strategy, international relations, computational geometry, and machine learning.

The paper ends with "The End"

Contents

1 Introduction

Throughout military history, survival has depended on the geometry of detection and lethality.

Ancient infantry formations depended on line-of-sight engagement. Naval warfare depended upon gun range. Aircraft warfare depended upon radar and missile envelopes.

Modern submarine warfare extends this logic into a dynamic geometric field in which mobile destruction zones interact continuously through time.

This paper studies the following question:

> Suppose N submarines move from origins to destinations. Each submarine destroys any other submarine that enters its destruction radius. What fraction of submarines survive after all submarines either arrive at their destinations or are destroyed?

The problem combines ideas from

- naval warfare,
- kinetic theory,
- stochastic geometry,
- graph theory,
- military operations research,
- computational geometry,
- large-scale agent systems,
- geopolitical deterrence theory.

2 Model

2.1 Origins and Destinations

Submarine i begins at

$$O_i = (x_i, y_i)$$

and travels toward

$$D_i = (c_i, d_i).$$

Its velocity is

$$U_i = (u_i, v_i).$$

The trajectory is

$$P_i(t) = (x_i + u_i t,\ y_i + v_i t).$$

The travel time is

$$T_i = \frac{\sqrt{(c_i - x_i)^2 + (d_i - y_i)^2}}{\sqrt{u_i^2 + v_i^2}}.$$

Submarine i exits the system at $t = T_i$.

2.2 Destruction Radius

Each submarine possesses radius

$$R_i.$$

Submarine i destroys submarine j whenever

$$\|P_i(t) - P_j(t)\| \leq R_i.$$

3 Pairwise Interception Theory

Define

$$\Delta x = x_i - x_j,$$

$$\Delta y = y_i - y_j,$$

$$\Delta u = u_i - u_j,$$

$$\Delta v = v_i - v_j.$$

The relative position is

$$r(t) = (\Delta x + \Delta u t, \Delta y + \Delta v t).$$

Therefore

$$\|r(t)\|^2 = At^2 + Bt + C,$$

where

$$A = \Delta u^2 + \Delta v^2,$$

$$B = 2(\Delta x \Delta u + \Delta y \Delta v),$$

$$C = \Delta x^2 + \Delta y^2.$$

A kill occurs whenever

$$At^2 + Bt + (C - R_i^2) \leq 0.$$

3.1 First Interception Time

The first interception time is

$$\tau_{ij} = \min\left\{t \geq 0 : At^2 + Bt + (C - R_i^2) = 0\right\}.$$

If no real solution exists, interception never occurs.

4 Directed Kill Graph

Definition 1. *The kill graph is*

$$G = (V, E),$$

where

$$V = \{1, \ldots, N\}.$$

There exists an edge

$$i \to j$$

if submarine i can destroy submarine j.

Each edge carries weight

$$\tau_{ij}.$$

The earliest destruction time of submarine i is

$$K_i = \min_j \tau_{ji}.$$

Submarine i survives iff

$$K_i > T_i.$$

Define

$$S_i = \mathbf{1}(K_i > T_i).$$

The survivor fraction is

$$F = \frac{1}{N} \sum_{i=1}^{N} S_i.$$

5 Computational Algorithm

The exact solution requires evaluation of all pairwise encounters.

5.1 Pseudo-Code

```
Input:
    Origins O_i
    Destinations D_i
    Velocities U_i
    Ranges R_i

For every pair (i,j):
    Compute interception time _ij

Sort all kill events by time

For each event:
    If both submarines alive:
        Destroy target

Count survivors

Return:
    F = survivors / N
```

5.2 Complexity

Pairwise calculations require

$$O(N^2)$$

operations.
Event sorting requires

$$O(N^2 \log N).$$

Thus

$$O(N^2 \log N)$$

is sufficient for exact simulation.

6 Relative-Motion Geometry

Consider two submarines.
Define

$$r_0 = O_i - O_j$$

and

$$w = U_i - U_j.$$

Then

$$r(t) = r_0 + wt.$$

The minimum distance between trajectories is

$$d_{\min} = \sqrt{\|r_0\|^2 - \frac{(r_0 \cdot w)^2}{\|w\|^2}}.$$

A lethal encounter occurs iff

$$d_{\min} \leq R.$$

7 Minkowski Sausage Geometry

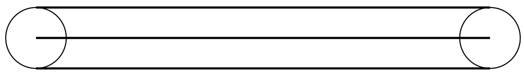
A submarine traveling distance L sweeps a lethal region

$$\mathcal{S} = \{x : \mathrm{dist}(x, \gamma) \leq R\}.$$

The swept area is

$$A_{\mathcal{S}} = 2RL + \pi R^2.$$

$$2RL + \pi R^2$$



This object is known in stochastic geometry as a Minkowski sausage.

8 Large-N Mean-Field Theory

8.1 Submarine Density

Let

$$\rho = \frac{N}{A}$$

denote force density.
Let

$$\bar{L} = E[L]$$

be the average path length.
The expected encounter count becomes

$$\Lambda = \rho(2R\bar{L} + \pi R^2).$$

8.2 Unit Square Result

For random endpoints in a unit square,

$$\bar{L} = \frac{2 + \sqrt{2} + 5\ln(1 + \sqrt{2})}{15}.$$

Numerically,

$$\bar{L} \approx 0.521405.$$

Hence

$$\Lambda = \rho(1.04281R + \pi R^2).$$

9 Poisson Encounter Approximation

In the thermodynamic limit, encounters become approximately independent.

Proposition 1. *The encounter count converges to a Poisson random variable.*

Thus

$$K \sim \text{Poisson}(\Lambda).$$

Therefore

$$P(K = k) = e^{-\Lambda}\frac{\Lambda^k}{k!}.$$

10 Main Survival Theorem

Theorem 1 (Large-N Submarine Attrition Law)**.** *Suppose*

1. *origins are independently distributed,*
2. *destinations are independently distributed,*
3. *velocities are bounded,*
4. *lethal radius equals R,*
5. *density converges to ρ.*

Then

$$F_N \to F_\infty = \exp\left(-\rho(2R\bar{L} + \pi R^2)\right).$$

Proof. The expected number of lethal encounters equals the density multiplied by the swept lethal area.

Thus

$$E[K] = \Lambda = \rho(2R\bar{L} + \pi R^2).$$

Under asymptotic independence, encounter counts converge to a Poisson law.
A submarine survives iff no encounter occurs.
Therefore

$$F = P(K = 0) = e^{-\Lambda}.$$

Substitution yields

$$F = \exp\left(-\rho(2R\bar{L} + \pi R^2)\right).$$

□

11 Monte Carlo Experiments

Simulations were performed for randomly distributed origins and destinations.

R	Estimated F
0.0025	0.3720
0.0050	0.1851
0.0100	0.0720
0.0200	0.0223
0.0400	0.0084

Table 1: Monte Carlo survival estimates.

The fitted law was

$$F(R) \approx 0.2749e^{-95.95R}.$$

The exponential structure agrees with the mean-field theory.

12 Statistical Interpretation

The model belongs to the family of

- random geometric graphs,
- spatial point processes,
- interacting particle systems,
- ballistic annihilation models.

The survival function is analogous to reliability theory:

$$S(t) = e^{-\lambda t}.$$

The lethal radius acts as an effective hazard multiplier.

13 Machine Learning Applications

The framework naturally supports:

1. multi-agent reinforcement learning,
2. autonomous naval routing,
3. adversarial planning,
4. stochastic battlefield simulation,
5. swarm optimization.

State vectors may contain

$$(X, Y, U, V, R).$$

Reward functions can be written as

$$J = \alpha(\text{survival}) + \beta(\text{mission completion}) - \gamma(\text{losses}).$$

14 Warfare Economics

Military survival directly affects economic output.
Let

$$M = \text{military capital}.$$

Expected surviving military capital is

$$E[M_{\text{surv}}] = MF.$$

Therefore

$$E[M_{\text{surv}}] = M \exp\left(-\rho(2R\bar{L} + \pi R^2)\right).$$

The attrition process behaves like depreciation of productive capital.

15 Political Science and Geopolitics

The model possesses implications for:

- deterrence stability,
- force dispersal,
- maritime chokepoints,
- strategic submarine basing,
- second-strike capability.

Increasing force concentration raises ρ and therefore decreases survivability. This creates a strategic trade-off between concentration and resilience.

16 International Relations

The framework provides a quantitative interpretation of naval deterrence.

A state may improve survivability by:

1. increasing geographic dispersion,
2. reducing detectable path overlap,
3. lowering encounter density,
4. increasing maneuver uncertainty.

These correspond to reductions in the effective encounter intensity

$$\Lambda.$$

17 Philosophical Implications

The model illustrates a broader principle:

> Survival is fundamentally geometric.

Whether biological, economic, political, or military systems persist depends upon the geometry of encounters.

The same mathematics appears in

- epidemiology,
- financial contagion,
- information diffusion,
- military attrition.

18 Conclusion

A complete mathematical framework for mobile submarine attrition has been developed.

The principal result is the asymptotic survival law

$$F_\infty = \exp\left(-\rho(2R\bar{L} + \pi R^2)\right).$$

This formula links geometry, density, and lethality into a single expression and provides a tractable approximation for large-scale naval conflict. The resulting theory connects stochastic geometry, warfare economics, international relations, military operations research, machine learning, and strategic deterrence into a unified analytical framework.

References

[1] W. Feller, *An Introduction to Probability Theory and Its Applications*, Wiley.

[2] J. F. C. Kingman, *Poisson Processes*, Oxford University Press.

[3] D. Stoyan, W. Kendall, J. Mecke, *Stochastic Geometry and Its Applications*, Wiley.

[4] M. Newman, *Networks*, Oxford University Press.

[5] J. D. Murray, *Mathematical Biology*, Springer.

[6] M. Osborne, A. Rubinstein, *A Course in Game Theory*, MIT Press.

[7] A. Washburn, *Search and Detection*, INFORMS.

[8] F. W. Lanchester, *Aircraft in Warfare*, Constable.

Glossary

F Survivor fraction.

R Lethal radius.

ρ Submarine density.

Λ Expected encounter count.

K Number of lethal encounters.

G Directed kill graph.

τ_{ij} Interception time.

$\bar{L}$ Average path length.

Minkowski Sausage Tubular neighborhood generated by a trajectory.

Poisson Approximation Large-N independent encounter limit.

The End